# Quizrr Chapter-Wise Test for JEE Main - 2025

**Q.1**

A vessel contains oil (density $= 0.8 \mathrm{gm/cm^3}$ ) over mercury (density $= 13.6 \mathrm{gm/cm^3}$). A homogeneous sphere floats with half of its volume immersed in mercury and the other half in oil. The density of the material of the sphere in $\mathrm{gm/cm^3}$ is

**(A)** $3.3$

**(B)** $6.4$

**(C)** $7.2$

**(D)** $12.8$

**Q.2**

In a hydraulic lift, compressed air exerts a force $\mathrm{F_1}$ on a small piston having a radius of $5\ \mathrm{cm}$. This pressure is transmitted to a second piston of radius $15\ \mathrm{cm}$. If the mass of the load to be lifted is $1350\ \mathrm{kg}$, find the value of $\mathrm{F_1}$ ? The pressure necessary to accomplish this task is

**(A)** $1.4 \times 10^5\ \mathrm{Pa}$

**(B)** $2 \times 10^5\ \mathrm{Pa}$

**(C)** $1.9 \times 10^5\ \mathrm{Pa}$

**(D)** $1.9\ \mathrm{Pa}$

**Q.3**

A concrete sphere of radius $R$ has a cavity of radius $r$ which is packed with sawdust. The specific gravities of concrete and sawdust are respectively $2.4$ and $0.3$ for this sphere to float with its entire volume submerged under water. Ratio of mass of concrete to mass of sawdust will be

**(A)** 8

**(B)** 4

**(C)** 3

**(D)** zero

**Q.4**

A cubical box is to be constructed with iron sheets $1\ \mathrm{mm}$ in thickness. What can be the minimum value of the external edge so that the cube does not sink in water? $\left[\delta_{\text{iron}} = 8000\ \mathrm{kg/m^3}, \delta_{\text{water}} = 1000\ \mathrm{kg/m^3}\right]$

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**(A)** $4.8\ \text{cm}$

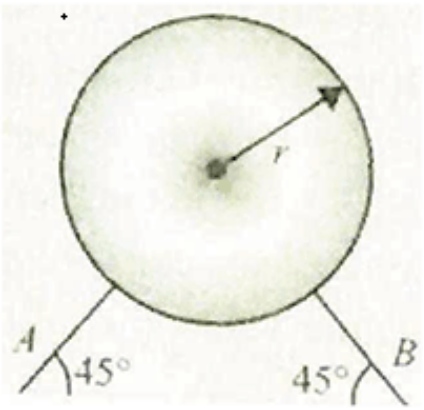
**(B)** $5.8\ \text{cm}$

**(C)** $6.9\ \text{cm}$

**(D)** $7.3\ \text{cm}$

**Q.5**

A hollow sphere of mass $\mathrm{M} = 50\ \text{kg}$ and radius $r = \left(\frac{3}{40\pi}\right)^{1/3}\ \mathrm{m}$ is immersed in a tank of water (density $\rho_{\mathrm{w}} = 10^3\ \text{kg/m}^3$ ). The sphere is tied to the bottom of a tank by two wires $\mathrm{A}$ and $\mathrm{B}$ as shown. Tension in wire $\mathrm{A}$ is $\left(\mathrm{g} = 10\ \text{m/s}^2\right)$




**(A)** $125\sqrt{2}\ \text{N}$

**(B)** $125\ \text{N}$

**(C)** $250\sqrt{2}\ \text{N}$

**(D)** $250\ \text{N}$

**Q.6**

In rising from the bottom of a lake, to the top, the temperature of an air bubble remains unchanged, but its diameter gets doubled. If $h$ is the barometric height (expressed in $\mathrm{m}$ of mercury of relative density $\rho$ ) at the surface of the lake, the depth of the lake is

**(A)** $8\rho\text{h metre}$

**(B)** $7\rho\text{h metre}$

**(C)** $9\rho\text{h metre}$

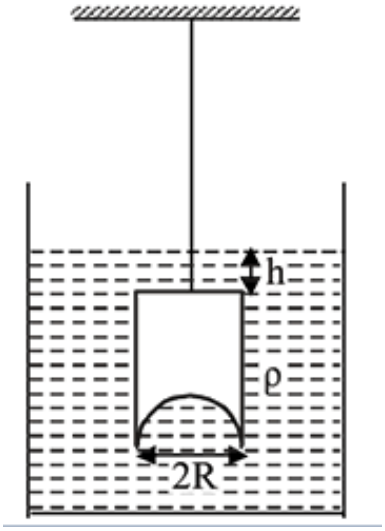
**(D)** $12\rho\text{h metre}$

**Q.7**

A hemispherical portion of radius $R$ is removed from the bottom of a cylinder of radius $R$. The volume of the remaining cylinder is $V$ and its mass $M$. It is suspended by a string in a liquid of density $\rho$ where it stays vertical. The upper surface of the cylinder is at a depth $h$ below the liquid surface. The force on the bottom of

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the cylinder by the liquid is




**(A)** $Mg$

**(B)** $Mg - V\rho g$

**(C)** $Mg + \pi R^2 h\rho g$

**(D)** $\rho g\left(V + \pi R^2 h\right)$

**Q.8**

An iceberg is floating in ocean. What fraction of its volume is above the water ? (Given : density of ice $=$ $900\ \mathrm{kg/m^3}$ and density of ocean water $=$ $1030\ \mathrm{kg/m^3}$ )

**(A)** $\frac{90}{103}$

**(B)** $\frac{13}{103}$

**(C)** $\frac{10}{103}$

**(D)** $\frac{1}{103}$

**Q.9**

A hemispherical bowl just floats without sinking in a liquid of density $1.2 \times 10^3\ \mathrm{kg/m^3}$. If outer diameter and the density of the bowl are $1\ \mathrm{m}$ and $2 \times 10^4\ \mathrm{kg/m^3}$ respectively then the inner diameter of the bowl will be

**(A)** $0.94\ \mathrm{m}$

**(B)** $0.97\ \mathrm{m}$

**(C)** $0.98\ \mathrm{m}$

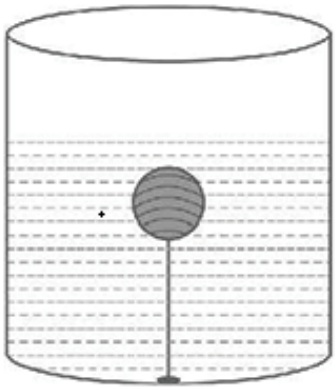
**(D)** $0.99\ \mathrm{m}$

**Q.10**

A solid sphere of density $\eta(>1)$ times lighter than water is suspended in a water tank by a string tied to its

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base as shown in fig. If the mass of thesphereis $m$, then the tension in the string is given by



**(A)** $\left(\frac{\eta-1}{\eta}\right)mg$

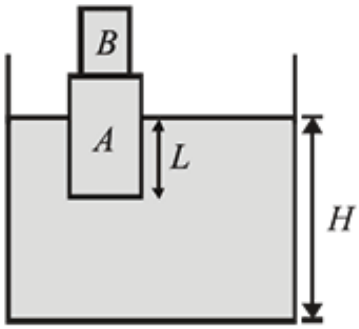
**(B)** $\eta mg$

**(C)** $\frac{mg}{\eta-1}$

**(D)** $(\eta-1)mg$

**Q.11**

Two wooden blocks $A$ and $B$ float in a liquid of density $\rho_{\mathrm{L}}$ as shown. The distance $\mathrm{L}$ and $\mathrm{H}$ are shown. After some time, block B falls into the liquid, so that $\mathrm{L}$ decreases and $\mathrm{H}$ increases. If density of block B is $\rho_{\mathrm{B}}$, find the correct option.




**(A)** $\rho_{\mathrm{L}}=\rho_{\mathrm{B}}$

**(B)** $\rho_{\mathrm{L}}>\rho_{\mathrm{B}}$

**(C)** $\rho_{\mathrm{L}}<\rho_{\mathrm{B}}$

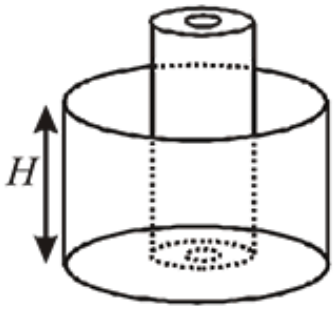
**(D)** unpredicatbale

**Q.12**

A hollow wooden cylinder of height $h$, inner radius $R$ and outer radius $2R$ is placed in a cylindrical container of radius $3R$. When water is poured into the container, the minimum height $H$ of the container for which

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cylinder can float inside freely is




**(A)** $\frac{h\rho_{\text{water}}}{\rho_{\text{water}}+\rho_{\text{wood}}}$

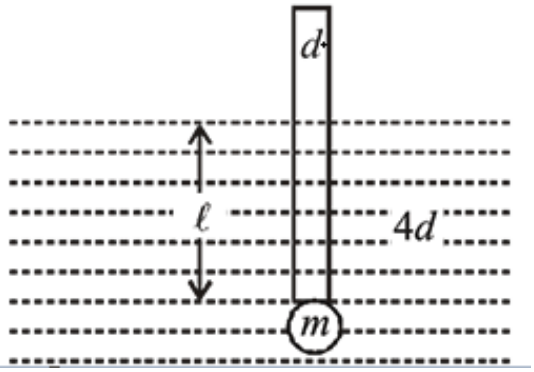
**(B)** $\frac{h\rho_{\text{wood}}}{\rho_{\text{water}}}$

**(C)** $h$

**(D)** $\frac{h^2}{R}$

**Q.13**

A uniform wooden stick of length $L$, crosssection area $A$ and density $d$ is immersed in a liquid of density $4d$. A small body of mass $\mathrm{m}$ and negligible volume is attached at the lower end of the rod so that the stick floats vertically in stable equilibrium then




**(A)** $m > dAL$

**(B)** $m < dAL$

**(C)** $m < dAL/2$

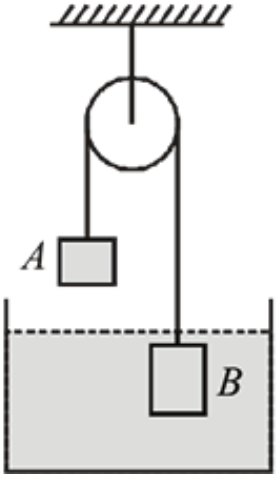
**(D)** $m < dAL/4$

**Q.14**

In the arrangement as shown, $m_B = 3m$, density of liquid is $\mathrm{r}$ and density of block $\mathrm{B}$ is $2\rho$. The system is released from rest so that block $\mathrm{B}$ moves up when in liquid and moves down when out of liquid with the same

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acceleration. Find the mass of block $A$.




**(A)** $\frac{7}{4}$ m

**(B)** 2 m

**(C)** $\frac{9}{2}$ m

**(D)** $\frac{9}{4}$ m

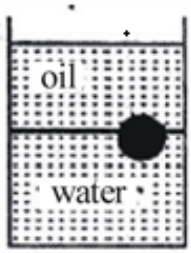
**Q.15**

$A$ ball is made of a material of density $\rho$ where $\rho_{\text{oll}} < \rho < \rho_{\text{watac}}$ with $\rho_{\text{oll}}$ and P wate representing the densities of oil and water, respectively. The oil and water are immiscible. If the above ball is in equilibrium in a mixture of this oil and water, which of the following pictures represents its equilibrium position?

**(A)**




**(B)**




**(C)**




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**(D)**




**Q.16**

A $0.5\ \text{kg}$ block of brass (density: $8 \times 10^3\ \text{kg m}^{-3}$ ) is suspended from a string. What is the tension in the string if the block is completely immersed in water?
$\left(g = 10\ \text{ms}^{-2}\right)$

**(A)** $5\ \text{N}$

**(B)** $\frac{0.5}{8\times10^3}\ \text{N}$

**(C)** $\frac{5}{8}\ \text{N}$

**(D)** $\left[5 - \frac{5}{8}\right]\ \text{N}$

**Q.17**

An object weighs $m_1$ in a liquid of density $d_1$ and that in liquid of density $d_2$ is $m_2$. The density d of the object is

**(A)** $d = \frac{m_2d_2 - m_1d_1}{m_2 - m_1}$

**(B)** $d = \frac{m_1d_1 - m_2d_2}{m_2 - m_1}$

**(C)** $d = \frac{m_1d_2 - m_2d_1}{m_1 - m_2}$

**(D)** $d = \frac{m_2d_1 - m_1d_2}{m_1 - m_2}$

**Q.18**

Two blocks $\text{A}$ and $\text{B}$ are made of different kinds of wood. Block $\text{A}$ floats in water with $\frac{1}{4}th$ of its above the surface of water. Block $B$ floats in water with $\frac{2}{3}rd$ of its volume below the surface of water. The ratio of the densities of $\text{A}$ and $\text{B}$ is

**(A)** $3 : 2$

**(B)** $5 : 3$

**(C)** $9 : 8$

**(D)** $4 : 3$

**Q.19**

The terminal velocity of a drop of water of radius $0.0015\ \text{mm}$, which is falling in air whose coefficient of

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viscosity is $2.0 \times 10^{-5}$ kg $m^{-1}$ $s^{-1}$, will be:
(The density of water $= 10^3$ kg $m^{-3}$ and g = 10 m $s^{-2}$ )

**(A)** $1.0 \times 10^{-4}\ \mathrm{m\ s^{-1}}$

**(B)** $2.0 \times 10^{-4}\ \mathrm{m\ s^{-1}}$

**(C)** $2.5 \times 10^{-4}\ \mathrm{m\ s^{-1}}$

**(D)** $5.0 \times 10^{-4}\ \mathrm{m\ s^{-1}}$

**Q.20**

Consider two syringes with diameters of the smaller piston and larger piston being 1 cm and 3 cm respectively, excluding needle. Both are filled with water and are connected with a tightly fitted rubber tube filled with water. Now, if a force of 10 N is applied to the smaller piston, then what is the force exerted on the larger piston?

**(A)** 30 N

**(B)** 60 N

**(C)** 90 N

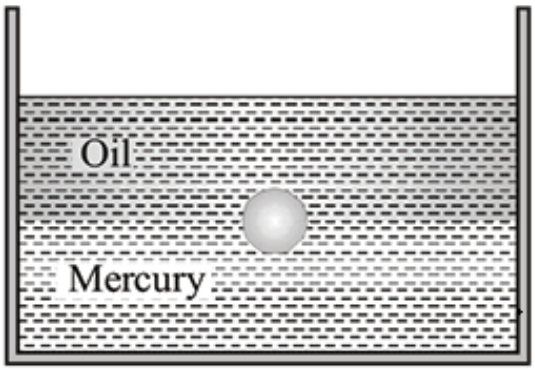
**(D)** 100 N

## Answers & Solutions

**Q.1 Answer:**

7.2

**Solution:**




**As the sphere floats in the liquid. Therefore its weight will be equal to the upthrust force on it Weight of sphere** $= \frac{4}{3}\pi R^3 \rho g \quad \ldots$

**Upthrust due to oil and mercury** $= \frac{2}{3}\pi R^3 \times \sigma_{\text{oil}}\, g + \frac{2}{3}\pi R^3 \sigma_{Hg} g \ldots$ **(ii)**

**Equating (i) and (ii)** $\frac{4}{3}\pi R^3 \rho g = \frac{2}{3}\pi R^3 0.8g + \frac{2}{3}\pi R^3 + 13.6g$

$\Rightarrow 2\rho = 0.8 + 13.6 = 14.4 \Rightarrow \rho = 7.2$

**Q.2 Answer:**

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$1.9 \times 10^5$ Pa

**Solution:**

**Since pressure is transmitted undiminished throughout the fluid (Pascal's law)** $\mathrm{F}_1 = \frac{\mathrm{A}_1}{\mathrm{A}_2}\ \mathrm{F}_2 = \frac{\pi(5\times5)}{\pi(15\times15)}(1350 \times 9.81)$

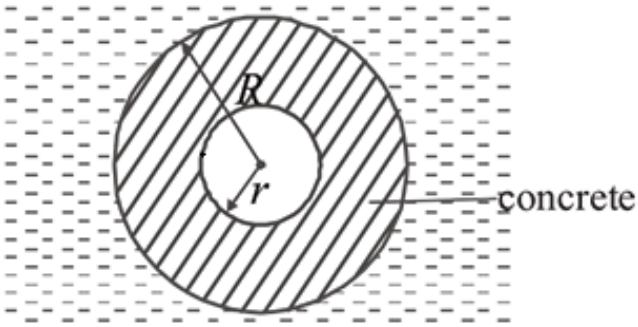
$\approx 1.5 \times 10^3$ N

**The air pressure that will produce this force** $\mathrm{P} = \frac{\mathrm{F}_1}{\mathrm{A}_1} = \frac{1.5\times10^3}{\pi(5\times10^{-2}\ \mathrm{m})^2} = 1.9 \times 10^5$ Pa

**Q.3 Answer:**

4

**Solution:**




$(m_{\text{concrete}} + m_{\text{sawdust}})\, g = V \rho_{\text{w}} g$

$2.4 \times \frac{4}{3}\pi\left(R^3 - r^3\right) + 0.3 \times \frac{4}{3}\pi r^3 = \frac{4}{3}\pi R^3 \times 1$

**or** $R^3 = \frac{3}{2}r^3$

$\therefore \frac{m_{\text{concrete}}}{m_{\text{sawdust}}} = \frac{\rho_{\text{concrete}}\ \frac{4}{3}\pi\left(R^3 - r^3\right)}{\rho_{\text{sawdust}}\ \frac{4}{3}\pi r^3}$

$= \frac{2.4}{0.3}\frac{\left(\frac{3}{2}r^3 - r^3\right)}{r^3} = 4$

**Q.4 Answer:**

4.8 cm

**Solution:**

**At minimum value of edge (let say a) the cube will just float in the water Then, volume of iron used**

$= \left(6\mathrm{a}^2\right)(\mathrm{t})$

$\mathrm{ATt} = 1\ \mathrm{mm} = 10^{-3}\ \mathrm{mm}$

**Volume of iron** $= 6\mathrm{a}^2 \times 10^{-3}$

**Mass of iron used = sensitivity of iron × volumes of iron**

$= 8000 \times 10^{-3} \times \left(6\mathrm{a}^2\right)$

$= 48\mathrm{a}^2$

**As the volume of cube** $= \mathrm{a}^3$

**volume of water displaced** $= \mathrm{a}^3$

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**To make the cube float**

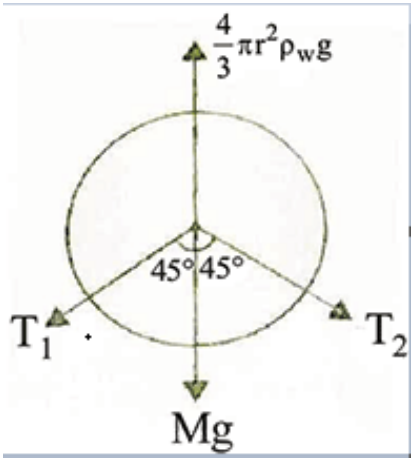
$$\text{Mass of iron in cube} = \text{Mass of water displaced}$$
$$\Rightarrow 48a^2 = 10^3 \times a^3$$
$$48 \times 10^{-3}\ \text{m} = \text{a}$$
$$\Rightarrow \text{a} = 48\ \text{mm}$$

**Q.5 Answer:**

$250\sqrt{2}\ \text{N}$

**Solution:**

$\text{T}_1 = \text{T}_2, \text{Mg} + \sqrt{2}\ \text{T}_1$
$= \frac{4}{3}\pi r^3 \rho_\text{w} g\ \text{T}_1$
$= 250\sqrt{2}\ \text{N}$




**Q.6 Answer:**

$7\rho\text{h}$ metre

**Solution:**

**Temperature constant** $\Rightarrow PV =$ **constant,**
**The volume is 8 times as diameter gets doubled. pressure gets** $1/8$ **times**
**air pressure*** $8 =$ **air pressure** $+$ **pressure due to water**
$=$ **height due to water** $= 7\rho\text{h}$

**Q.7 Answer:**

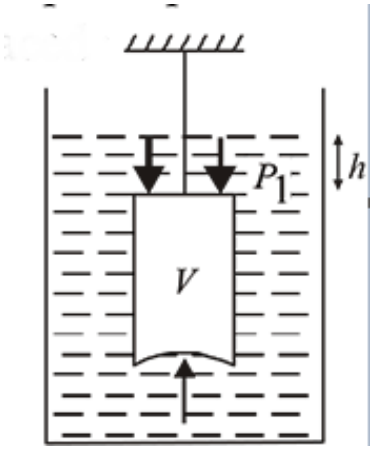
$\rho g\left(V + \pi R^2 h\right)$

**Solution:**

**According to Archimedes principle Upthrust = Wt. of fluid displaced** $F_\text{bottom} - F_\text{top} = V\rho g$
$\therefore \quad F_\text{bottom} = F_\text{top} + V\rho g$
$= P_1 \times A + V\rho g$
$= (h\rho g) \times \left(\pi R^2\right) + V\rho g$
$= \rho g\left[\pi R^2 h + V\right]$

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0. 0. $-0.8$




**Q.8 Answer:**

$\frac{13}{103}$

**Solution:**

**Let $V$ be the volume of iceberg and let $x$ be the fraction of volume above water. Using law of floatation, weight of floating body = weight of liquid displaced by part of the floating body inside the liquid. Therefore, $V\rho_{\text{ice}}\, g = (1-x)V\rho_{\text{water}}\, g$ Using the value of $\rho_{ice}$ and $\rho_{\text{water}}$ , we get $x = (13/103)$**

**Q.9 Answer:**

$0.98$ m

**Solution:**

**Weight of the bowl** $= \text{mg} = V\rho g = \frac{4}{3}\pi\left[\left(\frac{D}{2}\right)^3 - \left(\frac{d}{2}\right)^3\right]\rho g$

**Where** $\text{D}$ **= Outer diameter**

$\text{d}$ **= Inner diameter,** $\rho$ **= Density of bowl Weight of the liquid diplaced by the bowl** $= V\sigma g = \frac{4}{3}\pi\left(\frac{D}{2}\right)^3\sigma g$

**where $\sigma$ is the density of the liquid For the floation** $\frac{4}{3}\pi\left(\frac{D}{2}\right)^3\sigma g = \frac{4}{3}\pi\left[\left(\frac{D}{2}\right)^3 - \left(\frac{d}{2}\right)^3\right]pg$

$\Rightarrow \left(\frac{1}{2}\right)^3 \times 1.2 \times 10^3 = \left[\left(\frac{1}{2}\right)^3 - \left(\frac{d}{2}\right)^3\right]2 \times 10^4$

**By solving we get** $\text{d} = 0.98$ m.

**Q.10 Answer:**

$(\eta - 1)mg$

**Solution:**

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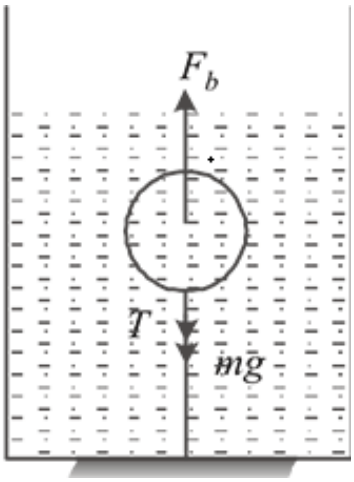
$$T + mg = F_b$$

$$\therefore \quad T = F_b - mg$$

$$= V\rho_w g - mg$$

$$= \frac{m}{(\rho_w/\eta)}\rho_w g - mg$$

$$= (\eta - 1)mg.$$




**Q.11 Answer:**

$\rho_L > \rho_B$

**Solution:**

**As block B falls, $L$ will decrease. Block B will displace, volume of liquid $(V_1)$ Equal to its own volume when it is in the liquid. $V_1 = V_B = \frac{M_B}{\rho_B}$**
**When block $B$ is on block $A$, it will displace the volume of liquid $(V_2)$ whose weight is equal to the weight of the block B. $V_2\rho_L g = M_B g = (V_1\rho_B)\, g$ or $V_2\rho L = V_1\rho_B$**
**Since $H$ increases, $V_1 > V_2$ So, $\rho_L > \rho_B$**

**Q.12 Answer:**

$\frac{h\rho_{wood}}{\rho_{water}}$

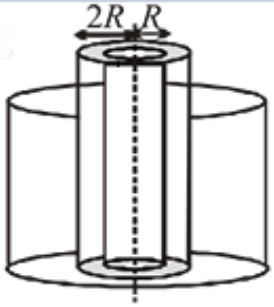
**Solution:**

$$mg = v\rho_{wood}\, g = \left(3\pi R^2 h\right)\rho_{wood}\, g$$

$$\left(3\pi R^2 h\right)\rho_{wood}\, g = \left(3\pi R^2 H\right)\rho_w g$$

$$\frac{h\rho_{wood}}{\rho_{water}} = H$$

**0**




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**Q.13 Answer:**

$m > dAL$

**Solution:**

**For equilibrium** $4dA\ell g = (dAL + m)g$

**and** $\frac{\ell}{2} > Y_{cm}$ **(for rotational equilibrium)** $Y_{cm} = \frac{m\times 0 + dAL(L/2)}{m+dAL} = \frac{dAL^2}{2(m+dAL)}$

$\ell > \frac{AL^2 d}{(ALd+m)} \Rightarrow \frac{m+dAL}{4dA} > \frac{AL^2 d}{(ALd+m)}$

$m > ALd$

**Q.14 Answer:**

$\frac{9}{4}$ m

**Solution:**

**Acceleration when block B is in the liquid,** $\mathrm{a}_1 = \frac{\mathrm{m_A g} - (\mathrm{m_B g} - \text{ upthrust })}{(\mathrm{m_A} + 3\ \mathrm{m})}$

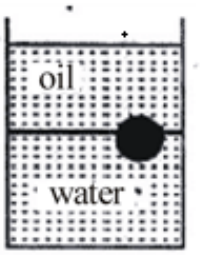
$= \frac{m_A g - \left(3mg - \frac{3m}{2\rho}\rho g\right)}{(m_A + 3m)} (\uparrow)$

**Acceleration when block** B **is outside if the liquid,** $a_2 = \frac{3mg - m_A g}{(m_A + 3m)} (\downarrow)$

**Given** $\mathrm{a}_1 = \mathrm{a}_2$**, we get** $\mathrm{m_A g} - \frac{3}{2}\mathrm{mg} = 3\mathrm{mg} - \mathrm{m_A g}$

$\Rightarrow \quad 2\ \mathrm{m_A g} - \frac{9}{2}\mathrm{mg} \Rightarrow \mathrm{m_A} = \left(\frac{9}{4}\right)\mathrm{m}$

**Q.15 Answer:**




**Solution:**

$p_{0\|} < p < p_{wnm}$

**Oil is the least dense of them so it should settle at the top with water at the base. Now the ball is: denser then oil but less denser then water. So, it will sink through oil but will not sink in water. So it will stay at the oil-water interface.**

**Q.16 Answer:**

$\left[5 - \frac{5}{8}\right]$ N

**Solution:**

**Volume of block** $= \frac{0.5}{8\times 10^5}\ \mathrm{m}^3$

**Volume of displaced water** $= \frac{0.5}{8\times 10^3}\ \mathrm{m}^3$

**Weight of displaced water**

$\frac{0.5}{8\times 10^3} \times 10^3 \times 10\ \mathrm{N} = \frac{5}{8}\ \mathrm{N}$

**This is equal to upward buoyant force.**

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$\left[5 - \frac{5}{8}\right]$ N

**Tension in string is actual weight less the upward buoyant force. So, tension is** $\left[5 - \frac{5}{8}\right]$ N

**Q.17 Answer:**

$d = \frac{m_1 d_2 - m_2 d_1}{m_1 - m_2}$

**Solution:**

**According to archimedes principle,**

**The weight of the given body in a liquid of density** $d_1$ **is**

$Vd - d_1 g = m_1 g$

**Where** $d, g, V$ **are the density of the object, acceleration due to gravity and volume of the object respectively.**

**The weight of the given body in a liquid og density** $d_2$ **is**

$Vd - d_2 g = m_2 g$

**On solving the two equations,**

$\frac{d - d_1}{d - d_2} = \frac{m_1}{m_2}$

$\therefore d = \frac{m_1 d_2 - m_2 d_1}{m_1 - m_2}$

**Q.18 Answer:**

9 : 8

**Solution:**

**Let** $V_1$ **be the volume of block** $A$ **and** $p_1$ **is density. Then, from the principle of flotation, we have**

$\rho_1 V_1 g = \left(1 - \frac{1}{4}\right) V_1 \times \rho \times g$

**or** $P_1 = \frac{3p}{4}$

**where** $\rho$ **is the density of water. Similarly for block B we have**

$\rho_1 V_2 g = \frac{2V_2}{3} pg$ **or** $p_2 = \frac{2p}{3}$

$\therefore \frac{P_1}{P_2} = \frac{3p}{4} \times \frac{3}{2p} = \frac{9}{8}$

**Q.19 Answer:**

$2.5 \times 10^{-4}$ m s$^{-1}$

**Solution:**

**Here,** $r = 0.0015$ mm $= 0.0015 \times 10^{-3}$m

$\eta = 2.0 \times 10^{-5}$ kg m$^{-1}$s$^{-1}$

$\rho 1.0 \times 10^{3}$kg m$^{-3}$

$g = 10$ m s$^{-2}$

**Neglecting the density of air, the terminal velocity of the water drop is**

$v_T = \frac{2}{9}\frac{r^2 \rho g}{\eta} = \frac{2 \times 0.0015010^{-3^2} \times 1.0 \times 10^3 \times 10}{9 \times 2.0 \times 10^{-5}}$

g $= 10$ m s$^{-1}$

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**Q.20 Answer:**

90 N

**Solution:**

**Since pressure is transmitted undiminished throughout the water**

$\therefore \frac{F_1}{A_1} = \frac{F_2}{A_2}$

**Where $F_1$ and $F_2$ are the forces on the smaller and on the larger pistons respectively and $A_1$ and $A_2$ are the respective areas.**

$$\therefore F_2 = \frac{A_2}{A_1} F_1 = \frac{\pi (D_2/2)^2}{\pi (D_1/2)^2} F_1 = \frac{D_2^2}{D_1} F_1$$

$$= \frac{\left(3 \times 10^{-2} \text{m}\right)}{\left(1 \times 10^{-2} \text{m}\right)^2} \times 10 \quad \text{N} = 90 \quad \text{N}$$

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